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Gauging the effects of the German COVID-19 fiscal stimulus package

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ABSTRACT

We simulate the fiscal stimulus package set up by the German government to alleviate the costs of the COVID-19 pandemic in a dynamic New Keynesian multi-sector general equilibrium model. We find that, cumulated over 2020–2022, output losses relative to steady state can be reduced by more than 6 PP. On average, welfare costs of the pandemic can be mitigated by 11%, or even by 33% for liquidity-constrained households. The long-run present value multiplier of the package amounts to 0.5. Consumption tax cuts and transfers to households primarily stabilize private consumption, and subsidies prevent firm defaults. The most cost-effective measure is an increase in productivity-enhancing public investment. However, it only fully materializes in the medium to long-term. Relative to the respective pandemic impact, some sectors such as the energy and the manufacturing sector benefit above average from the fiscal package, while the effect for some services sectors turns out to be below average.

1. Introduction

The COVID-19 pandemic and the resulting lockdown measures have caused a major economic recession around the world, and also weighed heavily on the German economy. To alleviate its impact, fiscal policy makers in Germany set up a large stabilization and stimulus package, including transfers to households and firms, easy access to publicly subsidized credit, tax relief and higher public (investment) spending. In this paper, we take a look at the core elements of these measures through the lens of a multi-sector New Keynesian dynamic stochastic general equilibrium model. To do so, we compare how key macroeconomic variables are expected to evolve after the COVID-19 pandemic (i.) without and (ii.) with the fiscal stimulus package. We also analyze the impact of each individual measure as well as fiscal multipliers and welfare implications.

Our model builds on Bouakez et al. (2018, 2021), featuring multiple interrelated production sectors that vary in their degree of price rigidity, factor intensities, use of intermediate inputs, and contribution to final demand. This setup allows us to analyze the heterogenous effects of the crisis and the policy responses across sectors. Since Germany is an export-oriented country and the COVID-19 crisis is characterized by interrupted global supply chains, we introduce international trade following Bergholt and Sveen (2014). Hence, the different sectors do not only trade goods with each other, but can also trade with firms abroad. As a simplification, we model Germany as a small open economy. The seven sectors included follow the NACE Rev. 2 classification and comprise (i.) agriculture, (ii.) mining, energy and water supply, (iii.) manufacturing, (iv.) a combination of trade, transporting, storage, accommodation and food service activities, (v.) IT and communication, (vi.) professional, scientific, technical, administrative and support service activities, and last but not least (vii.) art, entertainment and recreation, including other services activities.[1] To evaluate the impact of the crisis and the fiscal measures on firms' survival probabilities, we introduce a working capital channel and

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[1] For details on the statistical classification of economic activities in the European Community see Eurostat (2008).

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assume that firms can default on loans previously granted following Agnor et al. (2014). In our setup, sector-specific default rates differ, and so do the corresponding loan credit spreads. To analyze distributional effects on the household side, we choose a TANK (two-agent New Keynesian) model specification. Hence, we allow for liquidity-constrained consumers (also called rule-of-thumb (RoT) agents) and optimizers who own firms, banks and physical capital.

The economic effects are assessed by performing a conditional forecast based on the (projected) evolution of key macroeconomic and fiscal variables over the first year of the crisis until 2021:Q4. To obtain a baseline pandemic scenario without fiscal intervention, we keep the identified COVID-19 shocks and exclude the fiscal shocks. Our results can be summarized as follows.

First, our simulations confirm that the COVID-19 pandemic triggered a severe economic crisis. Aggregate output drops by around 18% relative to the steady-state when ignoring the fiscal intervention.[2] Consumption falls by around 15%, capital investment by 33% and hours worked by around 13%. In welfare terms, the average household in our model would be willing to give up about almost 1% of its steady-state consumption each period in order to avoid the COVID-19 recession.

Second, the impact of the crisis differs significantly across sectors. The manufacturing, cultural as well as trade, transport, accommodation and food services sector face the largest losses in output with up to 23%. Especially the cultural sector lastingly affected due to repeated and prolonged lockdown measures.

Third, the German fiscal stimulus package cannot prevent the initial drop in aggregate output, but mitigates the losses by about 6 PP cumulated over 2020–2023.[3] This corresponds to a present value output multiplier that reaches 0.3 in the second half of 2020 and converges to 0.5 in the long-run. The fiscal measures reduce welfare costs by around 11% on average, while those of liquidity-constrained households can be decreased by 33%.

Fourth, effects are heterogenous across fiscal instruments as well as across sectors and household types. In terms of sectoral differences, we find that relative to the sectoral drop in output that would have occurred without fiscal intervention, the mining, energy and water supply sector and the manufacturing sector benefit above average from the whole package, while the professional scientific and technical services as well as the cultural sector only benefit from the measures to a below-average extent. On the firm side, subsidies reduce default probabilities by essentially decreasing operating costs, which in turn reduces loan interest rates. On the household side, disposable income and consumption is stabilized by transfers as well as lower consumption and labor taxes. The fiscal measures increase the debt-to-GDP ratio by around 5 PP (to 72%) relative to a scenario without fiscal intervention. In terms of a cost–benefit analysis (by comparing the impact of the primary deficits of the measure and its cumulated impact on output), we find that increasing public investment is by far the most effective measure, which is driven by the assumption of private-sector productivity enhancing public capital. However, it fully materializes rather in the medium to long-term. The fiscal multiplier of the transfers is the second largest on impact as these heavily stabilize consumption of liquidity-constrained households, while subsidies yield the lowest multiplier. Moreover, transfers, consumption and labor tax cuts have in general the largest impact on the manufacturing and the trade, transportation & storage, accommodation and food services sectors.

Our paper is closely related to the literature studying the effects of economic policy responses to the current pandemic crisis, including, amongst others, Bayer et al. (2022), Elenev et al. (2022), Faria-e-Castro (2021), Kaplan et al. (2020), Hacıoğlu-Hoke et al. (2021), Pfeiffer et al. (2020) and Boscá et al. (2021). The first four papers analyze the impacts of various fiscal measures as a response to the COVID-19 crisis in the United States. Kaplan et al. (2020) use a HANK model that features more heterogeneity than the TANK specification we have chosen. They also find large and heterogenous welfare effects, also depending on the containment measures. Hacıoğlu-Hoke et al. (2021) report that low-income households have benefited most from government support in the UK. The latter two studies investigate the effects of short-term work allowances and liquidity guarantees in the European Union, and the stabilizing effects of the Spanish fiscal response to the Covid-19 crisis. Gourinchas et al. (2021) analyze the effects of fiscal policy in different countries, including advanced and emerging market economies, also focusing on the heterogeneity across sectors. Arellano et al. (2020) and Hürtgen (2020) address fiscal stability issues due to the pandemic, which we entirely ignore in this analysis (as they are, most likely, less important in Germany). Christl et al. (2022) analyze the effects of short-time working scheme and discretionary measures on household income, relying on a micro-level approach. They find the one-off payments to be a beneficial tool to support poorer households.

Baqaee and Farhi (2022) and Darougheh (2021) document that the pandemic affects different sectors differently. That government spending can have heterogenous effects on sectoral value added is documented in Bouakez et al. (2021). Hence, we think a multi-sectoral setup is essential when analyzing crises impacts and fiscal policy responses. Of course, the COVID-19 crisis is also a health crisis. Given that the multi-sector structure already adds considerable complexity in our model, we however refrain from explicitly combining an epidemic and an economic model (as in, for example, Atkeson (2020), and Eichenbaum et al. (2021)). Moreover, we neither contribute to the discussion about the extent to which the economic recession is driven by the pandemic itself or by lockdowns and how this might affect the impact of fiscal interventions (see, e.g., Auray and Eyquem, 2020). Nor do we touch upon the debate about the design of containment measures (see, e.g., Berger et al., 2022; Chari et al., 2020) or potential adverse economic side effects of lockdowns.

To the best of our knowledge, our paper is the first to focus on the effects of Germany's discretionary policy responses to the pandemic using a multi-sector dynamic general equilibrium model. It highlights which fiscal policy measure stabilizes (sectoral) output and household-type-specific consumption to what extent and, hence, gives guidance on how to set up potentially additional

[2] Note, that Chudik et al. (2021) finds similar values for Germany in a threshold-augmented global VAR analysis.

[3] Note, that we ignore additional effects from (extended) automatic stabilizers, equity injections and credit guarantees in our analysis due to the lack of reliable data and/or difficulties in implementing them into our model analysis. They are expected to have further contributed to output stabilization.

fiscal stimulus programs. Still, future research on the effects of the pandemic should also focus on the long-term consequences (as discussed,e.g., in Jordà et al., 2022) and the question of whether fiscal policy is capable of stabilizing the economy in the long-run, too (see also Priesmeier and Stähler, 2011).

The rest of the paper is organized as follows. In Section 2, we present the model. The calibration of the model is detailed in Section 3. The simulation design is laid out in Section 4. Section 5 discusses the results. Section 6 concludes.

2. The model

In this section, we lay out the structure of our multi-sector dynamic stochastic general equilibrium framework. Unless otherwise indicated, we express model variables in per-capita terms. Total population size is normalized to one.

2.1. Households

Following Gali et al. (2007b), we assume that the economy is populated by a share $\mu \in [0,1)$ of liquidity-constrained households (rule-of-thumb consumers), who do not participate in asset markets and consume their entire income each period, and a remaining share $(1-\mu)$ of capital and firm owners (optimizers). They are labeled by the superscript $i = o, r$ for optimizing and rule-of-thumb (RoT) households, respectively. The utility function of each type of representative household in group i at time t is given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{\left(\epsilon_t^C C_{i,t} - \kappa_{i,N} \epsilon_t^{\kappa_N} N_{i,t}^{\zeta} . X_{i,t}\right)^{1-\sigma} - 1}{1-\sigma} \right], \tag{1}$$

where $X_{i,t} = X_{i,t-1}^{1-\gamma^{ghh}} \cdot C_{i,t}^{\gamma^{ghh}}$, which makes preferences non-time-separable in consumption C_t^i and labor N_t^i of a type-i household. Such preferences shut off the wealth effect on the labor supply and are commonly used in open economy DSGE models. As outlined in Jaimovich and Rebelo (2009), these preferences nest as special cases the two classes of utility functions most widely used in the business cycle literature: If $\gamma^{ghh} = 1$ one gets preferences of the class discussed in King et al. (1988). If $\gamma^{ghh} = 0$ the preferences resemble those proposed by Greenwood et al. (1988). The parameter $\sigma \leqslant 0$ denotes the inverse of the elasticity of intertemporal substitution and $\kappa_{i,N}$ measures the relative weight of the disutility of labor. $\mathbb{E}_0$ is the expectations operator $\mathbb{E}_t$ at time $t = 0$, ϵ_t^C a consumption preference shock, and $\epsilon_t^{\kappa_N}$ a labor disutility shock, which we describe in detail later.

Optimizing households consume, supply labor, invest in physical capital, K_t, which pays a return r_t^k, buy public debt, B_t^o, and deposit interest-bearing savings, D_t^o, with banks to maximize their utility (1) for $i = o$ subject to the CPI-deflated real budget constraint

$$\begin{aligned} &\left(1+\tau_t^c\right) C_t^o + B_t^o + D_t^o + \frac{P_t^I}{P_t^C} I_t^o + \frac{P_t^I}{P_t^C} S\left(I_t^o, K_{t-1}^o\right) K_{t-1}^o + T_t^o = \left(1-\tau_t^w\right) w_t N_t^o \\ &+ R_{t-1}^d \frac{D_{t-1}^o}{\pi_t^C} + R_{t-1} \frac{B_{t-1}^o}{\pi_t^C} + \left[\left(1-\tau_t^k\right) r_t^k + \delta^k \tau_t^k\right] K_{t-1}^o + TR_t^o + \Pi_t^o + \Psi_t. \end{aligned} \tag{2}$$

While P_t^C denotes the consumer price index (CPI), which will be derived in detail later, $\pi_t^C = P_t^C / P_{t-1}^C$ is CPI inflation, w_t is the real wage rate, R_t^d represent real gross interest payments on outstanding deposits, R_t are real interest payments on outstanding public debt, TR_t^o (T_t^o) are per-capita lump-sum transfers (taxes), Π_t^o denote firm profits and Ψ_t are bank dividends stemming from the ownership of firms and banks. τ_t^c, τ_t^k and τ_t^w are consumption, capital income and labor income tax rates, respectively. Returns on physical capital net of depreciation allowances are taxed at rate τ_t^k. Moreover, we introduce convex capital accumulation costs of the form $S(I_t^o, K_{t-1}^o) = \kappa^I/2\left(I_t^o/K_{t-1}^o - \delta^k\right)^2$ (see, among others, Ireland, 2003). The law of motion for physical capital is given by:

$$K_t^o = (1-\delta^k) K_{t-1}^o + I_t^o. \tag{3}$$

We define $p_t^i = P_t^I / P_t^C$ as the relative price of investment in terms of consumption goods (because investment and consumption baskets may differ in our model). The optimization problem yields the standard first-order conditions.

RoT households also maximize (1) for $i = r$ with respect to labor supply subject to their real CPI-deflated budget constraint $\left(1+\tau_t^c\right) C_t^r = \left(1-\tau_t^w\right) w_t N_t^r + TR_t^r$. Economy-wide per-capita consumption and labor supply is calculated as $C_t = (1-\mu)C_t^o + \mu C_t^r$ and $N_t = (1-\mu)N_t^o + \mu N_t^r$, respectively. Aggregate transfers from the public to the household sector are given by $TR_t = (1-\mu)TR_t^o + \mu TR_t^r$. Household-specific variables for optimizers and rule-of-thumbers X_t^i are aggregated as $X_t = (1-\mu)X_t^o$ and $X_t = \mu X_t^r$, respectively.

Due to the multi-sector structure, we need to derive how much of aggregate consumption and investment spending $X_t \in \{C_t, I_t\}$ is channeled towards each sector $z \in Z$. We follow Bouakez et al. (2021) and assume that households' consumption and investment goods preferences across all Z sectors in our economy are given by

$$X_t = \left[\sum_{z=1}^{Z} \psi_{x,z}^{1-\sigma_x} X_{z,t}^{\sigma_x}\right]^{\frac{1}{\sigma_x}},$$

where $X \in \{C, I\}$. Moreover, $\psi_{x,z} \in [0,1]$ for $x \in \{c, i\}$ depicts the consumption utility value (for $x = c$) and the weight in the investment goods basket (for $x = i$) attached to a good produced in sector z. The parameter $\sigma_x \in (-\infty, 1)$ governs the elasticity of

substitution between these goods, which equals $1/(1-\sigma_x)$. It holds that $\sum_{z=1}^{Z}\psi_{x,z}=1$. Given $P_{z,t}^X$ as the consumer/investor price for goods of sector z, the household aims to minimize total consumption/investment spending

$$\min_{X_{z,t}} P_t^X\left[\sum_{z=1}^{Z}\psi_{x,z}^{1-\sigma_x}X_{z,t}^{\sigma_x}\right]^{\frac{1}{\sigma_x}}-\sum_{j=1}^{Z}P_{j,t}^X\epsilon_t^{X_z}X_{j,t},$$

which implies that the consumption/investment demand of sector-z goods is determined by

$$X_{z,t}=\psi_{x,z}\left(\frac{\epsilon_t^{X_z}P_{z,t}^X}{P_t^X}\right)^{-\left(\frac{1}{1-\sigma_x}\right)}X_t, \tag{4}$$

and the consumer/investor price index can be written as

$$P_t^X=\left[\sum_{z=1}^{Z}\psi_{x,z}\left(\epsilon_t^{X_z}P_{z,t}^X\right)^{-\frac{\sigma_x}{(1-\sigma_x)}}\right]^{-\frac{(1-\sigma_x)}{\sigma_x}}, \tag{5}$$

where $\epsilon_t^{X_z}$ is a cost shifter for sector-z goods that we use to describe the loss in demand for specific goods as a result of the pandemic (either because of preferences or because of policy measures, similar to the preference shifter described in Baqaee and Farhi, 2022, and Faria-e-Castro, 2021). As long as $\psi_{c,z}\neq\psi_{i,z}$ and/or $\sigma_c\neq\sigma_i$, it follows that $P_t^C\neq P_t^I$ and $p_t^i\neq 1$.

We additionally have to take into account that goods can be traded internationally. Hence, domestic consumers and capital investors not only need to decide which sectoral goods to purchase, but also whether to buy these goods at home or from abroad. We assume that households make this decision for each sectoral good separately. Specifically, we employ a CES aggregator for any sectoral good $X_{z,t}\in\{C_{z,t},I_{z,t}\}$:

$$X_{z,t}=\left[hb_{x,z}^{1-\sigma_{hb,x}}X_{z,h,t}^{\sigma_{hb,x}}+(1-hb_{x,z})^{1-\sigma_{hb,x}}X_{z,f,t}^{\sigma_{hb,x}}\right]^{\frac{1}{\sigma_{hb,x}}},$$

where $hb_{x,z}$ denotes the home bias for goods of sector z produced domestically, $\sigma_{hb,x}\in(-\infty,1)$ determines the elasticity of substitution between those goods produced domestically or abroad, $X_{z,h,t}$ denotes the domestically produced good and $X_{z,f,t}$ the corresponding good produced abroad. Given sectoral producer prices at home and abroad, $P_{z,h,t}^X$ and $P_{z,f,t}^X$, cost minimization of $P_{z,t}^X X_{z,t}-P_{z,h,t}^X X_{z,h,t}-\epsilon_t^{z,f}P_{z,f,t}^X X_{z,f,t}$ yields

$$X_{z,h,t}=hb_{x,z}\left(\frac{P_{z,h,t}^X}{P_{z,t}^X}\right)^{-\left(\frac{1}{1-\sigma_{hb,x}}\right)}X_{z,t},$$

and

$$X_{z,f,t}=(1-hb_{x,z})\left(\frac{\epsilon_t^{z,f}P_{z,f,t}^X}{P_{z,t}^X}\right)^{-\left(\frac{1}{1-\sigma_{hb,x}}\right)}X_{z,t} \tag{6}$$

as the consumption/investment demand for goods of sector z produced at home or abroad. When taking into account international trade, the corresponding consumer/investor price for goods of sector z is given by

$$P_{z,t}^X=\left[hb_{x,z}\,P_{z,h,t}^{X\,-\frac{\sigma_{hb,x}}{(1-\sigma_{hb,x})}}+(1-hb_{x,z})\left(\epsilon_t^{z,f}P_{z,f,t}^X\right)^{-\frac{\sigma_{hb,x}}{(1-\sigma_{hb,x})}}\right]^{-\frac{(1-\sigma_{hb,x})}{\sigma_{hb,x}}}. \tag{7}$$

Again, we allow for the home bias and the elasticity of substitution to differ across consumption and investment goods, and $\epsilon_t^{z,f}$ denotes a cost push shock on foreign goods governing losses in import demand due to the pandemic.

2.2. Labor and capital agencies

To determine the labor and capital supply that goes to each sector $z\in Z$ of our model economy, assume that perfectly competitive, representative labor and capital agencies hire the total amount of labor N_t at the real wage w_t and rent the total amount of capital K_t at the real rate r_t^k, selling it in turn to intermediate goods producers operating in Z different sectors (see also Bouakez et al., 2021). It is assumed that labor/capital input cannot perfectly move across sectors, so that

$$X_t=\left[\sum_{z=1}^{Z}\omega_{X,z}^{1-\nu_X}X_{z,t}^{\nu_X}\right]^{\frac{1}{\nu_X}},$$

for $X\in\{N,K\}$, where $\omega_{X,z}$ is the weight attached to labor/capital provided to sector z, and the elasticity of substitution of labor/capital across sectors can be derived from $\nu_X\in(-\infty,1)$. Hence, this parameter captures the degree of labor/capital mobility

between sectors. The labor agencies' optimization problem can be written as

$$\max_{X_{z,t}} \sum_{z=1}^{Z} x_{z,t} X_{z,t} - x_t \left[\sum_{z=1}^{Z} \omega_{X,z}^{1-\nu_X} X_{z,t}^{\nu_X} \right]^{\frac{1}{\nu_X}},$$

where $x_{z,t} = w_{z,t}$ and $x_t = w_t$ for $X = N$ and $x_{z,t} = r_{z,t}^k$ and $x_t = r_t^k$ for $X = K$, which leads to the following first-order condition characterizing the sector-specific demand for labor/capital:

$$X_{z,t} = \omega_{X,z} \left(\frac{x_{z,t}}{x_t} \right)^{-\left(\frac{1}{1-\nu_X}\right)} X_t. \tag{8}$$

By plugging this expression into the constant elasticity of substitution aggregator of labor/capital goods, we obtain the aggregate wage/capital rental rate

$$x_t = \left[\sum_{z=1}^{Z} \omega_{X,z} x_{z,t}^{-\frac{\nu_X}{(1-\nu_X)}} \right]^{-\frac{(1-\nu_X)}{\nu_X}}. \tag{9}$$

Again, we allow for $\nu_K \neq \nu_N$ and $\omega_{K,z} \neq \omega_{N,z}$, i.e., capital and labor substitutability as well as their weights may differ.

2.3. Firms

There is a continuum of producers that combine labor, capital, and a bundle of intermediate inputs to produce differentiated varieties of goods. We assume this to hold in each sector. These varieties are aggregated into a single good in each sector by a representative wholesaler who sells these to consumers and investors according to the consumption/investment baskets previously described.[4] We assume that intermediate goods producing firms face convex price adjustment costs following Rotemberg (1982).

Operating under perfect competition, the representative wholesaler in sector z maximizes profits

$$\max_{y_{z,h,t}(j)} P_{z,h,t} Y_{z,h,t} - \int_0^1 P_{z,h,t}(j) y_{z,h,t}(j) dj, \tag{10}$$

subject to the production technology $Y_{z,h,t} = \left(\int_0^1 y_{z,h,t}(j)^{(\theta_{p,z}-1)/\theta_{p,z}} dj \right)^{\theta_{p,z}/(\theta_{p,z}-1)}$, where $\theta_{p,z} > 1$ determines the elasticity of substitution between varieties in each sector, $y_{z,h,t}(j)$ is the retailer's demand for each differentiated input $j \in [0,1]$, and $P_{z,h,t}(j)$ is the producer price of each input. The first-order condition for the optimization problem is given by

$$y_{z,h,t}(j) = \left(\frac{P_{z,h,t}(j)}{P_{z,h,t}} \right)^{-\theta_{p,z}} Y_{z,h,t}, \tag{11}$$

capturing the demand for intermediate goods. By plugging the latter into the CES aggregator, we obtain $P_{z,h,t} = \left(\int_0^1 P_{z,h,t}(j)^{1-\theta_{p,z}} dj \right)^{1/(1-\theta_{p,z})}$, which is the producer price index for the home country.

Each intermediate goods producer $j \in [0,1]$ in sector z faces the technology

$$y_{z,h,t}(j) \leqslant \varepsilon_{z,t} \varrho_{z,t}(j) \; K_{t-1}^{g}{}^{\eta_{K^g,z}} \left(N_{z,t}(j)^{\alpha_{N,z}} K_{z,t-1}(j)^{1-\alpha_{N,z}} \right)^{\alpha_{H,z}} \left(H_{z,t}(j) \right)^{1-\alpha_{H,z}}, \tag{12}$$

where $\varepsilon_{z,t}$ is an aggregate productivity shock common to all sectors and $\varrho_t(j)$ is an idiosyncratic productivity uniform shock, distributed over the interval $(\underline{\varrho}, \overline{\varrho})$. The public capital stock is denoted by K_t^g, which is assumed to be productivity-enhancing along the lines of Leeper et al. (2010). The sector-specific parameter $\eta_{K^g,z} \geqslant 0$ determines how influential public capital is on sector-specific private production (see Bom and Ligthart, 2014, for a discussion).[5] The parameters $\alpha_{N,z} \in [0,1]$ and $\alpha_{H,z} \in [0,1]$ denote factor intensities, and $H_{z,t}(j)$ depicts intermediate goods needed in the production process.

We introduce a working capital channel following Agnor et al. (2014) and assume that firms need to pay wages, interest on capital input and intermediate goods purchases by relying on external financing. To do so, firm j has to take out a loan amounting to $L_{z,t}^f(j) = (1 + \tau_t^{sc}) w_{z,t} N_{z,t}(j) + r_{z,t}^k K_{z,t-1}(j) + P_{z,t}^H H_{z,t}(j)$. The respective loan rate to be paid is set before aggregate and idiosyncratic productivity shocks materialize. Then shocks are realized, production output becomes available, and employment, capital, intermediate-goods, loans, and prices adjust, taking interest rates and loan rates as given. We follow Stähler and Thomas (2012) and Gadatsch et al. (2016a), among others, and assume that firms have to pay social security contributions at rate τ_t^{sc} for all workers.[6] Along the lines of Agnor et al. (2014), some firms will default on their loans. The sector-specific interest rate for the

[4] An analogous basket is assumed for government consumption and investment spending as well as for exports (all of which we will derive in more detail below).

[5] We also ran the simulations assuming that public and private capital are connected via a CES aggregator along the lines of Coenen et al. (2013). The results remain pretty much unchanged.

[6] In Germany, social security payments are split between workers and firms (see Enders et al., 2020). We assume that the part borne by workers is included in τ_t^w, while the part borne by firms is given by τ_t^{sc}.

loans, $R^l_{z,t}$, will therefore depend on the default probability within this sector. Hence, the profit function of firm j in sector z in nominal terms is given by

$$P^C_t \Pi_{z,t}(j) = P_{z,h,t}(j) y_{z,h,t}(j) - R^l_{z,t} P^C_t L^f_{z,t}(j) + S_{z,t} - FC_z - \frac{\kappa^p_z}{2}\left(\frac{P_{z,h,t}(j)}{P_{z,h,t-1(j)}} - 1\right)^2 P_{z,h,t}(j) y_{z,h,t}(j), \tag{13}$$

where $S_{z,t}$ are subsidies received from government, and the last term on the right-hand side of this equation depicts quadratic price adjustment costs with $\kappa^p_z \geqslant 0$. FC_z are fixed costs in sector z.[7] Maximizing profits with respect to labor, capital and other intermediate inputs yields

$$R^l_{z,t} P^C_t w_t (1 + \tau^{sc}_t) = \alpha_{H,z} \alpha_{N,z} mc_{z,t} \frac{P_{z,h,t} y_{z,h,t}}{N_{z,t}}, \tag{14}$$

$$R^l_{z,t} P^C_t r^k_t = \alpha_{H,z}(1 - \alpha_{N,z}) mc_{z,t} \frac{P_{z,h,t} y_{z,h,t}}{K_{z,t-1}}, \tag{15}$$

$$R^l_{z,t} P^C_t P^H_{z,t} = (1 - \alpha_{H,z}) mc_{z,t} \frac{P_{z,h,t} y_{z,h,t}}{H_{z,t}}, \tag{16}$$

where we have dropped the index j as a result of symmetry and $mc_{z,t}$ denotes real marginal costs. Optimal prices are derived from

$$\beta E_t \left\{ \frac{\lambda^o_{t+1}}{\lambda^o_t} \kappa^p_z \left(\frac{P_{z,h,t+1}}{P_{z,h,t}} - 1\right)\left(\frac{P_{z,h,t+1}}{P_{z,h,t}}\right)^2 \left(\frac{Y_{z,h,t+1}}{Y_{z,h,t}}\right)\left(\frac{P^C_t}{P^C_{t+1}}\right)\right\} + 1 - \theta_{p,z} + (1 - \theta_{p,z}) mc_{z,t} = \kappa^p_z \left(\frac{P_{z,h,t}}{P_{z,h,t-1}} - 1\right)\frac{P_{z,h,t}}{P_{z,h,t-1}}. \tag{17}$$

A firm chooses to default whenever

$$\varrho_{z,t} < \frac{P^C_t R^l_{z,t} L^f_{z,t} + FC_z - S_{z,t} + \frac{\kappa^p_z}{2}\left(\frac{P_{z,h,t}}{P_{z,h,t-1}} - 1\right)^2 P_{z,h,t} y_{z,h,t}}{P_{z,h,t} \epsilon_{z,t} \; K^g_{t-1}{}^{\eta_{K^g}} \left(N^{\alpha_{N,z}}_{z,t} K^{1-\alpha_{N,z}}_{z,t-1}\right)^{\alpha_{H,z}} \left(H_{z,t}\right)^{1-\alpha_{H,z}}}. \tag{18}$$

This implies that an intermediate good producer declares default if its revenue after materialization of the idiosyncratic productivity shock is not high enough to cover its costs. The constraint may be alleviated by a government subsidy $S_{z,t}$. Also, whenever a firm defaults, its business is taken over by the remaining firms. The default threshold in sector z is thus given by $\bar{\varrho}_{z,t}$, which we obtain when Eq. (18) holds with equality.

In order to derive the sector-specific demand for intermediate inputs of sector z, we assume that it is determined by the CES aggregator,

$$H_{z,t} = \left[\sum_{\tilde{z}=1}^{Z} \psi^{1-\sigma_z}_{z,\tilde{z}} H^{\sigma_z}_{z,\tilde{z},t}\right]^{\frac{1}{\sigma_z}},$$

where $H_{z,\tilde{z},t}$ is the demand for intermediate good $\tilde{z}$ in sector z, with $\psi_{z,\tilde{z}}$ as the weight in the basket and $\sigma_z \in (-\infty, 1)$ determining the elasticity of substitution. In analogy to deriving sectoral consumption and investment demand $X_{z,t}$, we are able to derive $H_{z,\tilde{z},t}$. The same is true for the demand of domestic and foreign goods, where $hb_{z,\tilde{z}}$ denotes the home bias of sector z towards goods produced in sector $\tilde{z}$ and $\sigma_{hb,z} \in (-\infty, 1)$ representing the corresponding elasticity of substitution.

2.4. Banks

At the beginning of each period, the bank receives deposits from optimizers and gives loans to firms. Imposing that no reserves are required, the aggregate balance sheet of the bank is $(1-\mu)D^o_t = \sum_{z=1}^{Z} L^f_{z,t}$. We assume that the bank faces the refinancing (policy) rate R_t. No reserve requirement and perfect competition in the deposit market, both of which are assumed for reasons of simplicity, imply that $R^d_t = R_t$ (i.e., we abstract from a potential financial crisis and leave this for future research). Hence, optimizers receive the policy rate on their deposits from banks.

Next, we derive the sector-specific lending rate of loans granted to firms. The bank is exposed to default risk as firms' final output is subject to idiosyncratic shocks. Let $(1 - \chi_z)$, with $\chi_z \in (0, 1)$, be the fraction of firms' sales $P_{z,h,t}(j) y_{z,t}(j)$ that is lost in the event

[7] Fixed cost of production are introduced to reduce economic profits to zero in the steady state, thereby ruling out entry or exit (see Christiano et al., 2005).

of a default as in Agnor et al. (2014). Then, the bank's expected income from lending to a firm j in sector z is

$$EI_{z,t}(j) = \int_{\bar{\varrho}_{z,t}}^{\bar{\varrho}} R^l_{z,t}(j) L^f_{z,t}(j)\, f\left(\varrho_{z,t}\right) d\varrho_{z,t}$$
$$+ \int_{\underline{\varrho}}^{\bar{\varrho}_{z,t}} \left[\underbrace{\chi_z \frac{P_{z,h,t}(j)}{P^C_t} y_{z,t}(j) + \frac{S_{z,t}}{P^C_t} - \frac{FC_z}{P^C_t} - \frac{\kappa^p_z}{2}\left(\frac{P_{z,h,t}}{P_{z,h,t-1}} - 1\right)^2 \frac{P_{z,h,t}(j)}{P^C_t} y_{z,t}(j)}_{=\tilde{y}_{z,t}(j)} \right] f\left(\varrho_{z,t}\right) d\varrho_{z,t},$$

where $f\left(\varrho_{z,t}\right)$ is the density function for the idiosyncratic productivity shock $\varrho_{z,t}$. Whenever the idiosyncratic productivity shock exceeds the default threshold, the firm repays its loan. If it does not, the bank can seize the firms' remaining profits. However, $(1-\chi_z)P_{z,h,t}(j)y_{z,t}(j)$ is lost in this case. Using Eq. (18) and dropping the index j because of symmetry, we can re-write this as

$$EI_{z,t} = R^l_{z,t} L^f_{z,t} - \int_{\underline{\varrho}}^{\bar{\varrho}_{z,t}} \left[R^l_{z,t} L^f_{z,t} - \tilde{y}_{z,t}\right] f\left(\varrho_{z,t}\right) d\varrho_{z,t},$$
$$= \left[R^l_{z,t} - \rho^l_{z,t}\right] L^f_{z,t}, \tag{19}$$

where

$$\rho^l_{z,t} = \frac{1}{L^f_{z,t}} \frac{P_{z,h,t}(j)}{P^C_t} \varepsilon_z\, K^g_{t-1}{}^{\eta_{K^g}} \left(N^{\alpha_{N,z}}_{z,t} K^{1-\alpha_{N,z}}_{z,t-1}\right)^{\alpha_{H,z}} \left(H_{z,t}\right)^{1-\alpha_{H,z}} \left[1 - \frac{\kappa^p_z}{2}\left(\frac{P_{z,h,t}}{P_{z,h,t-1}} - 1\right)^2\right]$$
$$\cdot \int_{\underline{\varrho}}^{\bar{\varrho}_{z,t}} \left[\bar{\varrho}_{z,t} - \chi_z \varrho_{z,t}\right] f\left(\varrho_{z,t}\right) d\varrho_{z,t} \tag{20}$$

is the expected loss from lending to a defaulting sector-z firms. As we will see below, this expression turns out to be the finance premium charged on top of the policy rate when lending to firms. The bank's decision is thus to maximize profits by choosing how much to lend to each sector z. It does so by maximizing

$$\Pi^{bank}_t = \sum_{z=1}^{Z} R^l_{z,t} L^f_{z,t} - R_{t-1}(1-\mu) D^s_t \tag{21}$$

subject to the aggregate balance sheet described at the beginning of this section. Hence, the optimal interest on loans charged in sector z is given by

$$R^l_{z,t} = R_{t-1} + \rho^l_{z,t}, \tag{22}$$

reflecting that a firm's loan rate is determined before its aggregate and idiosyncratic productivity shocks materialize.

2.5. The government

The fiscal authority's budget constraint in CPI-deflated real terms is given by

$$B_t = R_{t-1} \frac{B_{t-1}}{\pi^c_t} + PD_t, \tag{23}$$

with

$$PD_t = \frac{P^G_t}{P^C_t} G_t + \frac{P^{I^g}_t}{P^C_t} I^g_t + TR_t - \tau^c_t C_t - \left(\tau^w_t + \tau^{sc}_t\right) w_t N_t$$
$$-\tau^k_t \left(r^k_t - \delta^k\right) K_{t-1} - (1-\mu) T^s_t, \tag{24}$$

as the primary deficit. The government must finance real government consumption and investment expenditures, $P^G_t/P^C_t G_t$ and $P^{I^g}_t/P^C_t I^g_t$, transfers to the household sector, TR_t, and interest payments on outstanding debt, $R_{t-1}B_{t-1}/\pi^c_t$, by taxing labor income (including social security contributions), capital returns and consumption, the issuance of new debt and lump-sum taxes. We follow Stähler and Thomas (2012) and Gadatsch et al. (2016b), among others, and assume full home bias in government consumption. This assumption is based on the observation that the import share in government consumption is, in general, significantly lower than in private consumption or investment (Brülhart and Trionfetti, 2001, 2004, and Trionfetti, 2000). The assumption is not crucial for our results as the modeled fiscal package does not include government consumption and the public investment part is comparably small. However, it has to be determined how fiscal authorities spend across sectors. We again assume a CES aggregator for public consumption and investment bearing in mind that no goods are purchased from abroad. This yields the common demand functions and price indices (analogous to those derived in the private sector) with the usual definitions for demand weights $\psi_{x,z}$ and σ_x in the public sector for $x \in \{G, I^g\}$. We assume that the public-sector capital stock evolves according to $K^g_t = (1-\delta^k)K^g_{t-1} + I^g_t$.

For $X \in \{G_t, I^g, TR, \tau^w, \tau^{sc}, \tau^c, \tau^k\}$, fiscal policy variables are assumed to be given by $X_t = \bar{X} + \epsilon^X_t$, where the bar indicates steady-state values and ϵ^X_t is a shock with a mean of zero. Hence, (most) fiscal variables are set to their steady-state values unless they are hit by a shock.

To guarantee stationarity, a fiscal rule along the lines of Schmitt-Grohé and Uribe (2007) and Kirsanova and Wren-Lewis (2012) is included in our model (see also Mitchell et al., 2000, for a discussion). We assume that this role is taken by the non-distortionary lump-sum tax levied on optimizers, which follows the process

$$\hat{T}_t^o = \rho^{fp}\hat{T}_{t-1}^o + \zeta^{debt}\hat{B}_{t-1}, \tag{25}$$

where $\hat{T}_t^o = \log\left(T_t^o/\bar{T}^o\right)$, $\hat{B}_t = \log\left(B_t/(\omega^{debt}Y_t)\right)$, ρ^{fp} is an autocorrelation parameter and ζ^{debt} a sensitivity parameter that determines how strongly the fiscal rule reacts to deviations of the public debt-to-GDP ratio from the target, ω^{debt}. Aggregate output, Y_t, is derived in the next section.

Regarding monetary policy, we assume that the central bank behaves accomodative over the simulation horizon, i.e. the nominal interest rate stays unchanged at steady state level.

2.6. International linkages and market clearing

In each sector z, product market clearing in nominal terms implies

$$\begin{aligned} P_{z,h,t}Y_{z,t} &= P_{z,t}^C C_{z,t} + P_{z,t}^I I_{z,t} + FC_z + TB_{z,t} + \sum_{\tilde{z}=1}^{Z} P_{z,h,t}H_{\tilde{z},z,t} \\ &+ P_{z,h,t}\left(G_{z,t} + I_{z,t}^g\right) + \frac{\kappa_z^p}{2}\left(\frac{P_{z,h,t}}{P_{z,h,t-1}} - 1\right)^2 P_{z,h,t}Y_{z,t}, \end{aligned} \tag{26}$$

where the sectoral trade balance (see also Bergholt and Sveen, 2014) is given by

$$TB_{z,t} = P_{z,h,t}\epsilon_t^{Exp,z}Exp_{z,t} - \epsilon_t^{z,f}P_{z,f,t}\left[C_{z,f,t} + I_{z,f,t} + \sum_{\tilde{z}=1}^{Z} H_{z,\tilde{z},t}\right]. \tag{27}$$

Given the small open economy assumption, exports, $Exp_{z,t}$, and prices for foreign goods, $P_{z,f,t}$, are assumed to be exogenously given. They may be shocked and follow an AR(1) process, where $\epsilon_t^{Exp,z}$ denotes the export shock.

Defining aggregate output as gross value added, we get

$$\begin{aligned} P_t^C Y_t &= \sum_{z=1}^{Z}\left(P_{z,h,t}Y_{z,t} - P_t^C P_{z,t}^H H_{z_t} - \frac{\kappa_z^p}{2}\left(\frac{P_{z,h,t}}{P_{z,h,t-1}} - 1\right)^2 P_{z,h,t}Y_{z,t} - FC_z\right) \\ &= P_t^C C_t + P_t^I I_t + P_t^G G_t + P_t^{Ig} I_t^g + P_t^C TB_t, \end{aligned}$$

where $P_t^C \cdot TB_t = \sum_{z=1}^{Z} TB_{z,t}$ and $\sum_{z=1}^{Z} P_t^C P_{z,t}^H H_{z_t} = \sum_{\tilde{z}=1}^{Z} P_{z,h,t}H_{\tilde{z},z,t}$. On the sectoral level, we define value added as the output available for consumption and investment (i.e. production less intermediate inputs as well as fix and adjustment costs). Given that we have a small open economy and, at the same time, assume a balanced trade balance per sector in each period, the policy rate equals the world interest rate $R_t = R$ (see also Bergholt and Sveen, 2014). Any exogenous shock follows an AR(1) process, which completes the model description.

The model is characterized by a set of nonlinear expectational difference equations. Once parameter values are assigned, the model can be linearized and solved using standard methods (see, e.g., Klein, 2000).

3. Calibration

In this section, we describe the steady-state calibration of the model. It consists of common parameters that describe the aggregate economy (mostly standard values from the literature) and sector-specific parameters that we need to describe the heterogenous production side and inter-sectoral linkages in our economy. The tables summarizing our parameter choices are related to Appendix. We describe these choices in more detail in this section.

Calibration is at a quarterly frequency. The discount factor is set to $\beta = 0.992$ to match an annual nominal interest rate of roughly 3.3%. The intertemporal elasticity of substitution is set to $\sigma_c = 1.5$, which is close to the mode estimates in Smets and Wouters (2003). We follow Jaimovich and Rebelo (2009) and set $\gamma^{ghh} = 0.001$ and $\psi = 2$. The parameter $\kappa_{i,N}$ is derived to obtain a labor share of $N_i = 0.333$. Capital depreciates at a rate of $\delta^k = 0.025$, see Cooley and Prescott (1995), implying an annual depreciation rate of 10%. Setting the share of liquidity-constrained consumers to 45% is in line with Ziegelmeyer et al. (2015). The capital adjustment cost parameter is set to 25, which is in line with estimates found in the literature (see, for example, Ireland, 2003). Substitution elasticities for goods produced in the different sectors are set as follows. With 0.5 for the (government) consumption and investment baskets, we choose similar values to the ones of Bouakez et al. (2021). For the intermediate inputs, we also follow Bouakez et al. (2021) and Atalay (2017), by setting a value of 0.1. Concerning the substitution elasticities of labor and capital, Bouakez et al. (2021) assume perfect substitutability. We do not go that far, but also assume a high substitutability and set the value to 5. As regards the substitutability between home and foreign (consumption/ investment/ intermediate) goods $\sigma_{hb,x}$, $x \in \{C, I, H\}$, we assume an equal value of 1/3 for all sectors, which corresponds to an elasticity of 1.5, a common value.

For the fiscal parameters, we rely on Gadatsch et al. (2016a), who estimate a fiscal DGSE model for Germany. Following Iacoviello (2005), we assume that, in steady state, optimizers and rule-of-thumb households consume the same. This is achieved by the government paying a transfer $\overline{TR}^r = 0.5775$ to liquidity-constrained households (the direct transfer to savers is assumed to be zero as

it would only increase the lump-sum tax accordingly; see also Gali et al., 2007b, and Forni et al., 2009, for a discussion). Targeting a debt-to-GDP ratio at an annual frequency of 60% (which is more or less the current value and corresponds to the Maastricht criteria), we get a lump-sum tax $\overline{T}^o = 0.549$ that closes the government budget constraint in steady-state. Following Schmitt-Grohé and Uribe (2007), we assume an autocorrelation parameter of 0.8 and a small debt reaction parameter of 0.005 for the fiscal rule.

The share of a bank's equity that is paid out to households amounts to $\nu^{bank} = 0.047$, which we derive endogenously from closing the bank's equity condition. We also derive the bank capital-to-loan ratio endogenously such that capital costs for banks are zero in steady-state. The adjustment cost parameter ζ^{bank} is set to 20 (Benes and Kumhof, 2015). All parameter values are in the range of values commonly chosen in the literature, and results are not very sensitive to changes in them.

For the multi-sector production part of our model, we distinguish between $Z = 7$ sectors, relying on the standard broad NACE classification[8] The first sector represents agriculture, forestry and fishing (A), the second is a conglomerate of mining and carrying, energy and water supply (B-D-E), and the third sector refers to manufacturing (C). The combination of trade, transporting, storage, accommodation and food service activities forms the fourth sector (G-H-I), followed by IT and communication (J), professional, scientific and technical as well as administrative and support service activities (M-N) and art, entertainment and recreation, including other services activities (R-S) as the last sector. Sticking to the NACE structure allows us to calibrate and match available disaggregated data. The calibration of the sector-specific parameters is mainly based on the most recent World Input Output Database (WIOD), providing sectoral data for the years 2000–2014 (see Timmer et al., 2015).[9] It encompasses data on socioeconomic accounts as well as input–output tables for 56 sectors and 43 countries.

The sectors in the model differ along several dimensions. First, they can be distinguished by their labor and capital supply, provided by the respective agency. The labor and capital weights, $\omega_{N,z}$ and $\omega_{K,z}$, respectively, are calculated based on WIOD's socioeconomic accounts. Specifically, we first add up the number of persons engaged and the nominal capital stock over all sectors, and then compute the sector-specific shares. Second, the technology of intermediate goods producers differs across sectors in the sense that we distinguish factor intensities for capital, labor and intermediate inputs as well as the productivity enhancement of public capital. The shares of intermediate inputs in total sectoral gross output $1-\alpha_{H,z}$ and the labor intensity implied by $\alpha_{N,z}$ match corresponding data within WIOD's input–output tables. By dividing the amount of intermediate inputs by gross output per industry, we can pin down the factor intensities for intermediate inputs, $1-\alpha_{H,z}$. The parameters $\alpha_{N,z}$ are then derived by dividing the share of compensation of labor in total sectoral gross output by $\alpha_{H,z}$.

The input–output tables also allow us to model detailed inter-sectoral trade shares of intermediate inputs, $\psi_{z,\tilde{z}}$, as well as the contributions to the final consumption, $\psi_{C,z}$, investment, $\psi_{I,z}$ and government consumption, $\psi_{G,z}$, goods. Due to the lack of data, we assume the shares of the public investment good $\psi_{IG,z}$ to be the same as for government consumption. The productivity of public capital is based on Bom and Ligthart (2014), whose meta-regression yields an average output elasticity of 0.106, while core public capital (such as roads, railways and so on) yields a larger short-run estimate of 0.131. Therefore, we assume that the professional scientific and technical services sector is the one that profits most from public capital.

Concerning the international dimension, we compute the home biases with respect to consumption $hb_{C,z}$, investment $hb_{I,z}$ and intermediate goods $hb_{z,\tilde{z}}$ by aggregating WIOD data of the remaining 42 countries to one *Rest of the world* counterpart of Germany and determine the respective shares of goods produced abroad and at home.

Furthermore, the sectors differ in price setting and default probabilities. Mark-ups θ_z^p for the different sectors stem from Christopoulou and Vermeulen (2012) and the fix cost is thus set to ensure zero profits in equilibrium. The sector-specific price durations are set as in Bouakez et al. (2021) and translated into Rotemberg price adjustment costs parameters κ_z^p along the lines of Keen and Wang (2007). The default probabilities from which we then extract the default threshold $\bar{\varrho}_z$ rely on business closing rates provided by Destatis for 2017.[10] Table A.6 in the Appendix compares actual to model implied moments along the lines of Born and Pfeifer (2014). The latter ones show correct signs and are similar in magnitude compared to the actual data as well, which corroborates our parameter and modeling choices.

4. Simulation design

In this section, we describe the simulation design of our analysis. The general idea is to match selected economic time series from 2020:Q1-2021:Q4 by different structural shocks. Specifically, in the first step, we include various types of COVID-19 as well as fiscal shocks. In the second step, we keep the identified COVID-19 shocks of the first step, but exclude the fiscal shocks to obtain a baseline pandemic scenario (without fiscal intervention) that can be compared to the scenario with fiscal intervention.

Concerning the nature of the COVID-19 shock, there seems to be consensus in the growing body of economics literature that it represents a combination of demand and supply shocks (see e.g. Eichenbaum et al., 2021, Pfeiffer et al., 2020, Bayer et al., 2022, Guerrieri et al., 2022, Brinca et al., 2021, and Balleer et al., 2020). Baqaee and Farhi (2022) show that the decomposition of the shock also differs across sectors which especially holds for the cultural sector, for example. Hence, we also use different shocks to replicate effects of the pandemic on selected variables. The endogenous variables we match comprise sectoral output, sectoral exports and imports (for which we can only distinguish between goods and services), aggregate consumption, hours worked and

[8] Note that we exclude the construction sector (F), financial and real estate activities (K-L), the public sectors (O-Q) as well as activities of households as employers; undifferentiated goods- and services-producing activities of households for own use (T) and activities of extraterritorial organizations and bodies (U).

[9] Our steady state calibration refers to mean values over this period. In order to extract and aggregate WIOD data, we set up a quite flexible calibration routine that allows for a custom choice of years, country and sector specifications.

[10] See https://www-genesis.destatis.de/genesis//online?operation=table&code=52111-0011&bypass=true&levelindex=0&levelid=1601033242933

wage income. All corresponding time series are part of Germany's national accounts published by Destatis in February 2022.[11] Table B.1 summarizes the matched paths of non-fiscal variables over the first two years of the crisis.

Regarding the shocks that are meant to reflect the pandemic, we differentiate between six types. First, a sector-specific shock to the costs of consumption ($\varepsilon_t^{C,z}$) reduces consumption demand, which may differ per sector. During the lockdown, consumption was either prohibited by regulations or more costly because of the containment measures. Second, we include a general consumption preference shock (ε_t^{C}) that is meant to capture peoples' behavior of postponing consumption due to increased uncertainty. Third, a labor disutility shock ($\varepsilon_t^{\kappa_N}$) reduces aggregate labor supply. It can also be interpreted as a proxy for people staying at home to avoid risks of infection while commuting, or rules restricting contacts at work. Fourth, a common shock to total factor productivity, mirroring possible restrictions concerning the production process. Fifth, we allow for sector-specific exports shocks ($\varepsilon_t^{Exp,z}$) that are supposed to match the fall in exports shown in the national accounts. An analogous sector-specific shock on import prices ($\varepsilon_t^{f,z}$) is used to reduce imports.[12] The intention here is to mirror, for example, the international dimension of the crisis with disrupted supply chains and a general reduction in world trade and travel.[13]

The German government has responded to the COVID-19 crisis with various expansionary fiscal measures. In terms of amounts, we rely on (projected) values of the Bundesbank .[14] In total, the fiscal measures included amount to 2.7% and 2.2% of trend GDP in 2020 and 2021, respectively, and 0.2–0.3% in the following two years. Hence, we observe a large (COVID-19 related) expansionary fiscal stimulus between 2020 and 2023 of about 5%. Roughly 65% of it comes from the actual fiscal stimulus package adopted in June 2020. The remaining part contains the block of measures that were established at the beginning of the crisis.

Regarding the intra-annual profiles, we assume that the first block of measures is split across the second to fourth quarter with weights of 0.6, 0.3 and 0.1, respectively. We choose to fade out the weights across the quarters in order to reflect the emergency assistance character of (some of) these measures. The fiscal stimulus package (from June) is split equally across the third and fourth quarters in the simulation. The intra-annual profiles in 2021–2023 are also assumed to be flat. Note that we only consider measures that affect the national accounts' deficit (except for tax deferrals). Hence, we do not include equity injections and credit guarantees, which affect public debt but not the deficit. These measures may have a significant additional stabilization impact (also see Bundesbank, 2020a,b).[15]

The German fiscal stimulus can be split into what is advertised by the government as "emergency assistance" and as the actual "fiscal stimulus package". Table 1 provides further details on the amounts and how the fiscal measures are implemented in the model. Two additional figures in the Appendix help to visualize the budget impact and timing. Fiscal emergency assistance measures primarily support firms to which almost 90% of the fiscal stimulus aimed at. Transfers to small and medium-sized businesses, and to the self-employed, make up the largest share. Direct transfers to theaters and other cultural institutions are also included (which we model as a sector-specific transfer to sector 7). Additionally, support was provided to firms by shifting the tax burden over time. Since these measures are intended to provide liquidity support, we model them as firm subsidies, too. Moreover, on the household side, social transfers other than in kind were expanded by more generous unemployment benefits, compensation of loss of earnings due to child care and higher replacement rates for short-time working arrangements. Furthermore, a bonus payment for people employed in the nursing sector was agreed upon.

Roughly 70% of the actual fiscal stimulus package is aimed at households. The VAT reduction accounts for the largest share, followed by a bonus transfer of 300 euro per child and tax reliefs for single parents. Firms, on the other side, benefited mainly from direct assistance, investment grants and subsidies. Income tax-related measures, such as an increased threshold for tax loss carrybacks and degressive depreciation rates for moveable assets, are further targeting a reduction of business costs in order to provide liquidity. We model them as subsidies to firms. Furthermore, the government promised to keep the rate for social security contributions below 40% until the end of 2021 by providing federal subsidies, which we model as a reduction in the contribution rate relative to the steady-state. Additionally, the fiscal stimulus package includes a public investment component with a focus on child care facilities, school renovations and constructions. It mainly consists of projects already planned but shifted earlier in time such that implementation delays should be less of a problem here. Concerning the distribution of transfers and subsidies across sectors, we use steady-state shares in aggregate output due to the lack of more detailed information.

To find the structural shocks needed to match the constrained paths of the endogenous variables previously described, we apply a conditional forecast. This is done using the reduced-form first-order state-space representation of the model $y_t = Ty_{t-1} + R\varepsilon_t$,

[11] See https://www.statistischebibliothek.de/mir/receive/DEHeft_mods_00138413.

[12] As pointed out by a referee, our modeling choice implies that sector-specific exports and import prices vary exogenously. For the counterfactual analysis, in which fiscal shocks are turned off, to be valid, requires to assume that exports and import prices are independent of domestic fiscal stimulus. This might be seen as a strong assumption. In a robustness exercise we keep the exports and import prices constant in the conditional forecast. We find our results to be unaffected by this procedure.

[13] Note, that all shocks are used to match all of the targeted variables at the same time. There is no one-to-one match of shocks and variables. For example, matched fluctuations in hours worked are not only a result of the labor disutility shock, but also due to the demand side shocks which lower firm's labor demand. Note, that we do not target sectoral price movements. In the Appendix, we depict model-implied price movements and actual ones.

[14] For an overview of the fiscal measures see Bundesbank (2021). We use updated values from April 2022. Note that we do not include measures earmarked specifically for health-related support such as expenses for additional intensive care beds, additional financial resources for hospitals, and other expenses for pandemic containment since we do not model a separate health care sector. As part of the fiscal stimulus package, the government also announced a large future investment package of 1.7% of GDP including programs tackling digitalization, climate change and health care. However, since details on the timing of implementation are lacking, and since the package is supposed to have a longer-run impact, we do not include it in our analysis.

[15] Moreover, there are other expansionary fiscal measures in place that are not related to the COVID-19 crisis, like the abolition of the solidarity surcharge. In the present paper, we only focus on COVID-19 related measures.

Table 1
Composition of the German COVID-19 fiscal stimulus measures.

Stimulus measure	2020	2021	2022	2023	Fiscal instruments
Emergency assistance					
Measures aimed at households:					
Transfers	0.1	0.1	0.0	0.0	TR
Measures aimed at firms:					
Liquidity support	1.2	0.3	0.0	0.0	S
Indirect taxes	0.0	0.1	0.0	0.0	τ^c
Fiscal stimulus package					
Measures aimed at households:					
Transfers	0.1	0.0	0.0	0.0	TR
Direct taxes	0.2	0.1	0.0	0.0	τ^w
Indirect taxes	0.7	0.4	0.0	0.1	τ^c
Measures aimed at firms:					
Liquidity support	0.3	1.0	0.2	0.2	S
Social security contributions	0.1	0.0	0.0	0.0	τ^{sc}
Public investment	0.1	0.1	0.0	0.0	I^g

Notes: Fiscal stimulus measures are expressed as a percentage of trend GDP. Note, that with the exception of tax deferrals, we only consider measures that affect the national accounts' deficit. Automatic stabilizers, equity injections and credit guarantees are not included.

where y_t includes pre- and non-predetermined variables, and ε_t are the structural shocks just described. The representation can be split into a controlled and uncontrolled part and then solved algebraically for the controlled shocks. Intuitively, the algorithm finds a combination of fiscal and non-fiscal shock sequences that matches the previously described entered time series. Note that only the controlled variables of the first eight quarters are fixed. Starting with the ninth quarter (2022:Q1), all variables move freely, driven by the shock propagation. Note, that the estimated non-fiscal shocks also reflect automatic stabilizers as these are not modeled explicitly since the focus is on discretionary measures. The same holds true for additional accommodation stemming from unconventional monetary policy measures like the Pandemic Emergency Purchase Programme of the ECB adopted at the end of March 2020. Hence, the "true" pandemic shocks might be underestimated with our approach.

5. Results

In this section, we first discuss the macroeconomic developments of selected key macroeconomic variables in the baseline pandemic scenario and in the one including fiscal measures. Then, we derive multipliers and analyze the welfare effects.

5.1. Macroeconomic effects of the crisis

The shocks describing the COVID-19 pandemic significantly reduce private consumption and hours worked as well as net exports. Higher import prices also increase production costs by driving up the prices for intermediate inputs. Unit labor costs rise, which also pushes up producer prices. This results in a significant fall in aggregate output and capital investment (the latter further dampening aggregate demand). German international competitiveness deteriorates,[16] and consumer price inflation increases. According to our model simulations, the German debt-to-GDP ratio increases from 60% in the initial steady-state to over 70% because of the pandemic (even without any fiscal intervention). A detailed figure visualizing these effects is relegated to Appendix.

The effects of the COVID-19 shocks also differ across sectors, as can be seen in Fig. 1. While most of the sectors share a large drop in output in the second quarter of 2020, the agricultural sector as well as the IT and communication sector perform relatively well during the crisis. The latter most likely benefitted from the crisis due to the increased use of teleworking and virtual working. The sectors hit hardest at the trough are manufacturing, arts, entertainment and recreation as well as the services sectors four and six with drops in output of −23%, −21% and −16%, respectively. Increases in unit labor costs and drops in hours worked are also pronounced for these sectors. The pandemic hit the manufacturing sector hard in the beginning through disrupted supply chains and a general reduction in world trade. Still, we observe a relatively quick recovery of output of the manufacturing sector, driven also by a come back of exports, while many other sectors experience negative rebounds in 2021 due to further lockdown measures.[17] The manufacturing sector shows a large decrease in capital investment, which also substantially impacts on aggregate capital investment, since manufacturing makes up for the largest share in the investment good bundle with almost 70% (see calibration table in the

[16] Note that this may be a result of assuming Germany to be a small open economy. A drop in export demand accompanied by a smaller drop in imports (as described in the previous section) is associated by a fall in international competitiveness in such a model. As the pandemic is a global shock, this must not be the case this time, and deteriorations in the value chain may change exports and imports as described without reducing competitiveness. However, given our model, in which competitiveness is an endogenous variable, we cannot do differently when matching trade data.

[17] Note that we cannot distinguish between trade, transport and accommodation and food services due to the lack of data. While trade and transport (except for passenger transport) recovered quickly in the third quarter of 2020, hotels and gastronomy certainly suffered larger losses comparable to the art, entertainment and recreation sector.

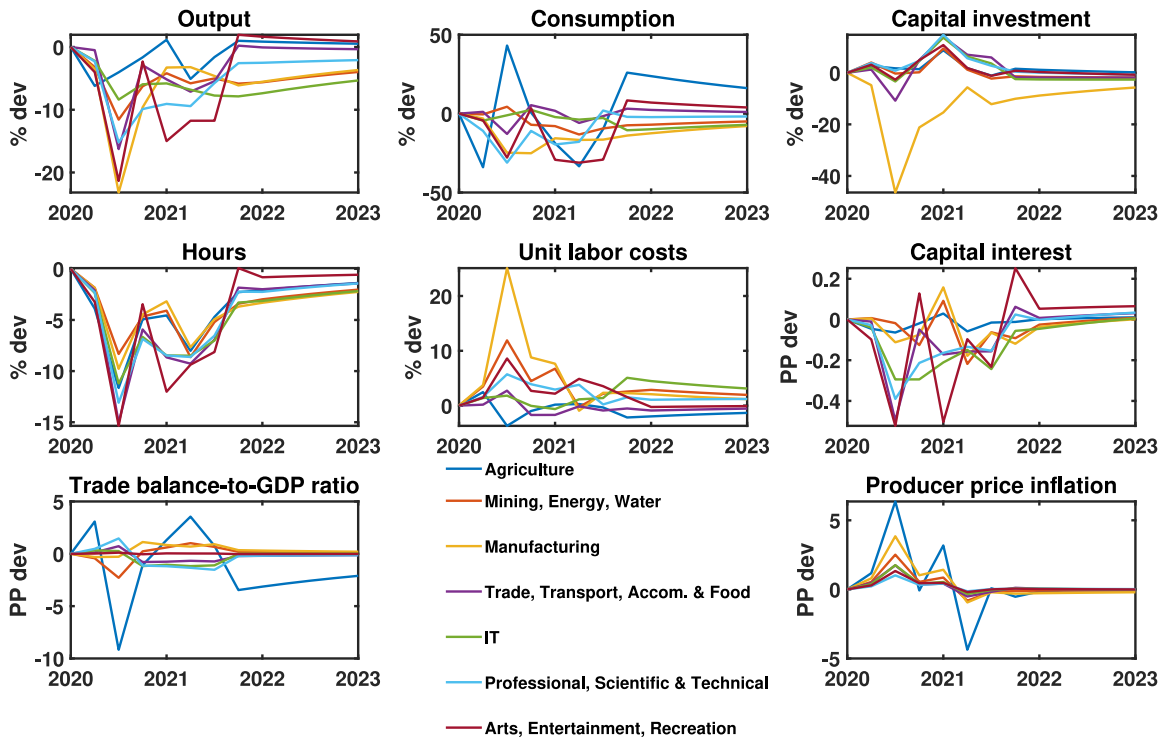

Fig. 1. Sectoral implications of COVID-19 pandemic. ***Notes:*** Figure plots (projected) paths of selected sectoral macroeconomic variables without fiscal intervention (baseline pandemic scenario) for each sector as indicated by the colored lines. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Appendix). The increase in production costs and the fall in demand have led to higher firm default rates (see figure C.3 in the Appendix). This applies in particular to the manufacturing sector (sector 3) and the cultural sector (sector 7), with the latter being hit the hardest and longest. Our simulations suggest that default rates in the former sector increase by over 15 PP. In the latter case, there is a double peak due to the second lockdown in 2021 with default rates increasing by about 6 PP.

5.2. Macroeconomic effects and channels of the fiscal measures

The fiscal package helps to mitigate the negative impact of the pandemic (see Fig. 2). While it cannot prevent the large drop in output in the second quarter of 2020, it stabilizes consumption and reduces unit labor costs for firms from the third quarter onward and throughout 2021. The effect on consumption amounts to over 2 PP at the peak. Capital investment is increased by over 1 PP. Together, this leads to a stabilization of output by about 1.5 PP in the second half of 2020. The debt-to-GDP ratio rises steadily by almost up to 5 PP in 2022. Cumulated over the first 3 years, the fiscal package stabilizes output by 6 PP, as can be seen from Fig. 3. Looking at the distributive effects, it turns out that the fiscal actions substantially support liquidity-constrained households' consumption with an cumulated effect of 10 PP, which is almost two times larger than the cumulated effect on optimizer's consumption.

What are the mechanisms behind these results? Fig. 3 plots the contribution of each fiscal measure of the entire fiscal stimulus to (cumulated) developments in output, investment and household type-specific consumption relative to a scenario without fiscal interventions. The following points can be noted here:

First, the reduction in consumption taxes and higher transfers to households increase disposable income. This stabilizes consumption demand, especially for rule-of-thumb consumers, who spend their entire income each period. However, higher debt-financed transfers to all households crowd out consumption demand of optimizing households. The reason is that higher transfers to rule-of-thumb households reduces their labor supply (given that consumption and leisure are normal goods). To increase production (in relative terms), real wages must increase. This reduces firm profits and income of optimizers, which overcompensates the increase in transfers devoted to them.

Second, as subsidies for firms stabilize firm income per period, they allow a larger fraction of firms to cover their fixed costs. This significantly reduces the probability of default, and default-implied output losses decrease. At the same time, lower probabilities of default imply lower loan interest rates. This is how firms' operating costs fall (due to the working capital channel). Lower default

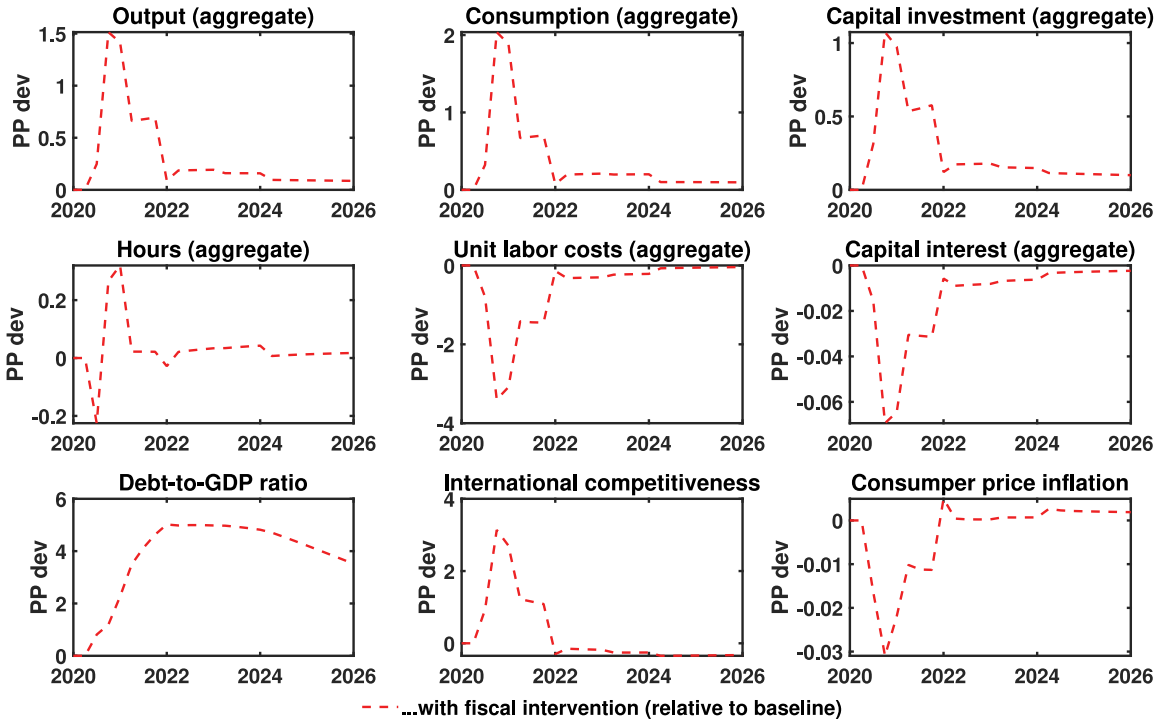

Fig. 2. Paths of key macro variables with fiscal stimulus packages relative to baseline scenario (aggregate). ***Notes:*** Figure plots (projected) paths of selected aggregated macroeconomic variables with the fiscal package (red dashed line) relative to the baseline scenario without fiscal intervention.

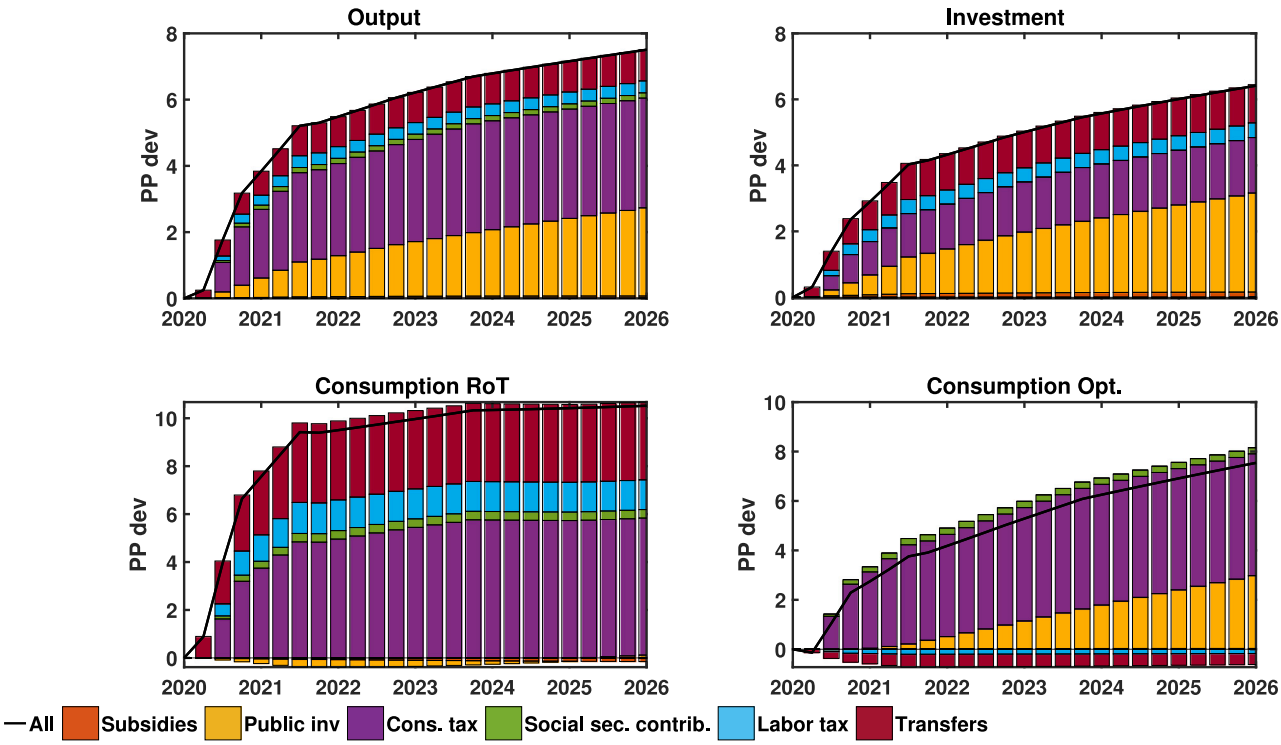

Fig. 3. Cumulated impact of fiscal measures on key macroeconomic variables (relative to baseline). ***Notes:*** Figure plots (projected) impact of different fiscal measures on selected macroeconomic variables. The contribution of each fiscal measure to these developments is shown by the colored bars as indicated. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

rates and operating costs stabilize wage income of households because marginal labor productivity does not fall as much.[18] This fosters demand.

In addition, lower labor income taxation augments net wage income, and lower social security contributions levied on firms reduces operating costs further. The reduction in labor income taxation has two output-increasing effects. First, it directly raises net wage income of households, which fosters demand. Second, because of the tax reduction, households are willing to accept lower gross wages (as they primarily care about their net labor income). The reduction in gross wages has a cost-dampening effect for firms, which are willing to increase employment and reduce prices in relative terms. The latter additionally fosters demand. The former increases the marginal product of capital and, thus, stimulates investment. Analogous effects stem from the reduction in the social security contributions, implying a decrease in firms' labor costs. Because households demand higher gross wages when social security contributions fall (given that wages reflect, in some way, the share of firm profits workers obtain), labor income also rises by this measure and households increase consumption (see also Attinasi et al., 2019, and Enders et al., 2020, for an in-depth discussion).

The most effective measure in the longer-run is the increase in public capital investment. It fosters output on impact through an increase in aggregate demand as investment goods are bought by the government from the private sector. Given that public investment increases the public capital stock, which positively affects private-sector productivity, private investment demand starts to increase eventually. The positive policy-induced "productivity" shock generates effects that are similar to a pure technology shock (see also Leeper et al., 2010, and Gadatsch et al., 2016a, for a more detailed discussion). As the productivity effect starts to kick in, the positive impact of the increase in public investment rises (see Fig. 3). It is noteworthy that the size of the effect very much depends on the productivity-enhancing impact of public capital, captured by the parameter $\eta_{K^g,z}$.[19]

In terms of a cost–benefit analysis, and given the parametrization of the impact of public capital on productivity, public investment appears to be a very effective tool. However, it takes some time before these effects materializes, which makes public investment possibly less effective in terms of short-term demand stabilization. Transfers as well as reliefs with regard to labor income taxation and social security contributions also seem to be relatively cost-efficient. The reduction in consumption taxation, on the contrary, turns out to be quite costly. However, it also affects output positively due to the stabilizing effect on liquidity-constrained households' consumption. While subsidies for firms are also costly, they do not show a large stabilizing effect on output.

Sectoral winners and losers. The sectoral effects of the fiscal package are heterogeneous (see figures C.5–C.11). Output effects are particularly large in the mining, energy and water supply, the manufacturing as well as in the transportation, accommodation and food sector with over 2 PP at the peak, while output of the professional, scientific and technical services increases only slightly through the fiscal intervention. We note that the stabilization impact is mainly via consumption, while capital investment is affected to a lesser extent. Labor costs can be reduced noticeably in most sectors (except for agriculture, which is only modestly affected by the crisis).

As regards the effect of subsidies to firms, it is worthwhile to have a closer look at the evolution of the sectoral default probabilities, which fall below their steady-state levels in several sectors (see figure C.3). A default probability below the steady-state level indicates that, because of the fiscal package, firms that would have left the market in normal times now survive (for longer) even though a severe negative shock is hitting the economy. Together with the fact that subsidies make up a fairly large part of the fiscal costs, while having only a modest effect on output, this might be an indicator that too many subsidies were provided, at least to some sectors (according to our model simulations) on the one hand.[20] On the other hand, default probabilities converge again quickly, and an appropriate distribution of subsidies is difficult to assess in practice.

We use Fig. 4 to assess which sectors benefit most from the overall fiscal intervention in 2020. In the figure, we plot the sectoral output improvements relative to their sectoral output losses without fiscal intervention over the baseline output losses. The linear regression trend line is given in orange. Sectors above (below) the trend line benefit more (less) than average. The fact that the regression line is rather flat indicates that the fiscal intervention was not very well targeted, i.e. sectors that suffer more due to the crisis do not receive more fiscal support. We find that the mining, energy and water supply sector, the manufacturing as well as the trade, transport & storage, accommodation and food sector experience the largest output improvement relative to the baseline scenario. They benefit disproportionately compared with the professional, scientific and technical services and the cultural sector, which show the largest baseline output losses but a significantly smaller relative improvement. These sectors come off worst compared to the others. This might be due to the fact the output of the professional, scientific and technical service sector is purchased to a comparatively small extent as intermediate goods from other sectors (see table A.4). Hence, positive spill over effects do not play a major role in this sector.[21]

[18] The initial drop in hours worked relative to a scenario where only the pandemic is simulated (see Fig. 2) can be explained by the fact that the fiscal package stabilizes consumption. As consumption and leisure are normal goods, households also want to enjoy more leisure (work less) for any given wage. Whenever consumption stabilization fades out, hours worked start to increase again.

[19] If the productivity-enhancing impact is zero, we only observe a small positive impulse on impact (due to higher demand) that fades out quickly. If we differentiate productivity by sector (or assume that a single sector does not benefit from the type of public investment undertaken), different sectors would also be affected very differently by this shock (see also Bom and Ligthart, 2014, for a discussion). Hence, the extent to which public investment positively affects aggregate and sectoral output depends decisively on the degree of productivity-enhancement we assume. An important avenue for future research would be to address which sectors benefit from which type of public investment.

[20] Note that these results also depend on the simulated amounts of subsidies as well as on the specific distribution across sectors. Concerning the latter, we have assumed a distribution according to the steady-state shares (as discussed above), which may not represent the actual amounts of subsidies received by each sector.

[21] The figure for the situation in 2021 is relegated to Appendix. Now, the cultural sector lie on the trend line (which is now also steeper), but they still experience large output losses in the baseline scenario.

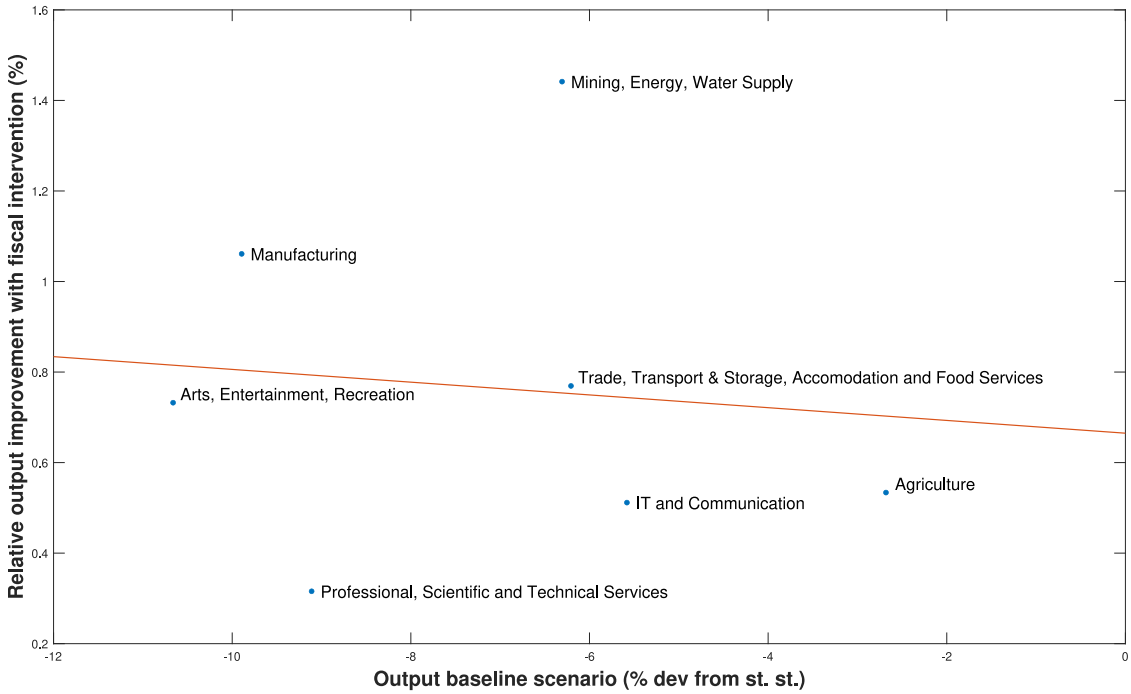

Fig. 4. Sectoral output improvements relative to baseline output losses (2020). ***Notes:*** Figure plots sectoral relative output improvements with fiscal package 2 over sectoral output losses in the baseline pandemic scenario. The yearly result for 2020 was computed by averaging over the respective four quarters. The orange line denotes the linear regression trend line.

5.3. Fiscal multipliers

We calculate the present value fiscal multipliers during the COVID-19 pandemic using a modified version of the multiplier presented by Leeper et al. (2010) and Bouakez et al. (2021) amongst others. To be precise, we first calculate the sum of deviations of output with the fiscal change relative to a situation without the fiscal change, $\left(Y_{t+i}^{fiscal} - Y_{t+1}^{pandemic}\right)$. Then, we divide this by the sum of the discounted fiscal change. The government transfer multiplier, for example, is then given by

$$PV_t(k) = \frac{\sum_{i=0}^{k} \bar{R}^{-i} \cdot \left(Y_{t+i}^{fiscal} - Y_{t+i}^{pandemic}\right)}{\sum_{i=0}^{k} \bar{R}^{-i} \cdot \left(TR_{t+i} - \bar{TR}\right)}, \tag{28}$$

where k indicates the time span for which the multiplier is calculated ($k = 0$ is the impact multiplier, $k = 4$ the year-one multiplier, and so on). The multiplier is calculated analogously for other fiscal instruments and for sectoral output. To obtain the multiplier of the entire package, we have to sum the measures in the denominator and take the corresponding transition path in the numerator.[22]

Fig. 5 shows the present value fiscal multipliers for output, consumption (for each household type) and investment over time. The output multiplier reaches a value of 0.3 in the second half of 2020, declines shortly afterwards and converges to 0.5 in the long-run. The consumption multipliers confirm our finding that the fiscal stimulus measures are quite effective in stabilizing consumption of liquidity-constrained households. However, the investment multiplier is rather small, driven by the fact that the share of public investment, which is usually associated with a multiplier larger than one, in the total package is also relatively small. Moreover, as shown in Fig. 3, some measures lead to a crowding-out of private investment, reducing the investment multiplier. Fig. 6 presents fiscal output multipliers across the various instruments of the package. Clearly, public investment yields the largest multiplier. It especially pays off in the long-run due to the assumption that public capital enhances also private capital's productivity, but also has the largest multiplier in the short-run with over 1. The second most effective instrument in terms of short-run stabilization are transfers with a multiplier of almost 0.8 on impact, followed by consumption and labor tax/ social security contribution cuts with corresponding multipliers of 0.64 and about 0.35, respectively. As shown in Table C.1, these measures have large consumption (RoT) multipliers. Moreover, transfers, consumption and labor tax cuts are found to be effective in stimulating output of the manufacturing sector (sector 3), and of trade, transportation & storage, accommodation and food services (sector 4). Both sectors have large shares in the consumption good bundle, such that they benefit in particular from measures stabilizing consumption (see Table A.3). Moreover,

[22] As the pandemic shocks are the same in both scenarios, this is close to calculating the fiscal stimulus package starting in the steady-state (see also Boysen-Hogrefe et al., 2020, Box 2, for a discussion).

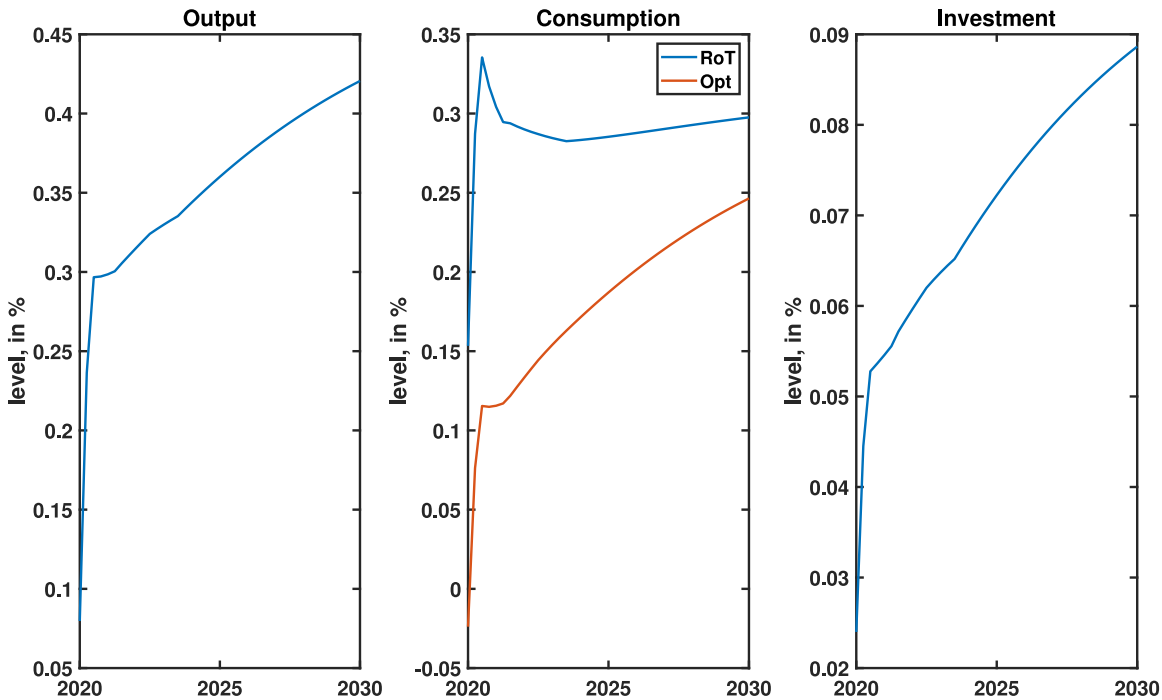

Fig. 5. Present value fiscal multipliers. ***Notes:*** Figure plots present value fiscal multipliers for output, consumption by household types (Rule of thumbers and optimizers) and investment, respectively.

they benefit by the labor tax reduction due to their large labor weights. In contrast, subsidies yield very small multipliers across all variables. The fact that these make up the largest share of the whole fiscal package also explains the rather small model-implied overall fiscal output multiplier, which falls within the range of results from similar analyses, see e.g. GD, 2020, chapter 5).

There are two remarks to be made here. As described in Section 2.5, we assume that only lump-sum taxes levied on optimizers do the job of consolidating the fiscal stimulus in our simulations (as, for example, in Coenen et al., 2013). On the one hand, it is not clear whether or not such a non-distortionary consolidation instrument exists. If we were to use other fiscal instruments to bring the debt-to-GDP level back to the initial steady-state, the multiplier would decrease with the degree of distortions this instrument introduces in the economy. On the other hand, it is not entirely clear if the pandemic also affects the growth path in the long-run (which is suggested by some commentators; see Jordà et al., 2022). If it leads to a lower growth path, and if fiscal policy was able to prevent this (or at least to reduce the drop in long-run growth), multipliers could also be larger (see also Priesmeier and Stähler, 2011, for a more detailed discussion of such mechanisms). While these aspects of the pandemic are certainly of interest and quite important, we will leave them for future research.

5.4. Welfare

To measure the welfare effects of the fiscal stimulus program, we calculate the consumption-equivalent welfare gain from consumption, ce^i, for $i = o, r$, such that

$$\sum_{t=0}^{\infty} \beta^t U\left[(1+ce^i)\bar{C}^i, \bar{N}^i\right] = \sum_{t=0}^{\infty} \beta^t U\left(C_t^i, \bar{N}^i\right),$$

where the exact utility function $U(\cdot)$ is given by Eq. (1). Utility positively depends on the level of consumption, C_t^i, and negatively on hours worked, N_t^i (which measures forgone leisure). The bar indicates initial steady-state values. Hence, ce^i represents the amount of initial steady-state consumption a household of type i is willing to give up in order to live in a scenario without the shocks and the fiscal packages that we simulate. Results are summarized in Table 2. Note that aggregate economy-wide welfare is calculated as $ce = (1-\mu)\cdot ce^o + \mu \cdot ce^r$.

The pandemic is, as we can see, a costly event. To avoid it, households would be willing to give up over 0.7% of their steady-state level of consumption each period. Costs are higher for optimizing households with a magnitude of 1.12%. The reason is that the pandemic generates a huge loss in profits and capital gains, which is entirely borne by optimizers. The fiscal stimulus package set up by the government is able to reduce welfare costs by about 11%. In relative terms, rule-of-thumb households benefit more, with a reduction in welfare costs of 33% versus a reduction of about 7% for optimizers. However, bearing in mind that the typical costs of whole business cycles amount to roughly 1% of steady-state consumption, as discussed in Gali et al. (2007a) and Imrohoroglu

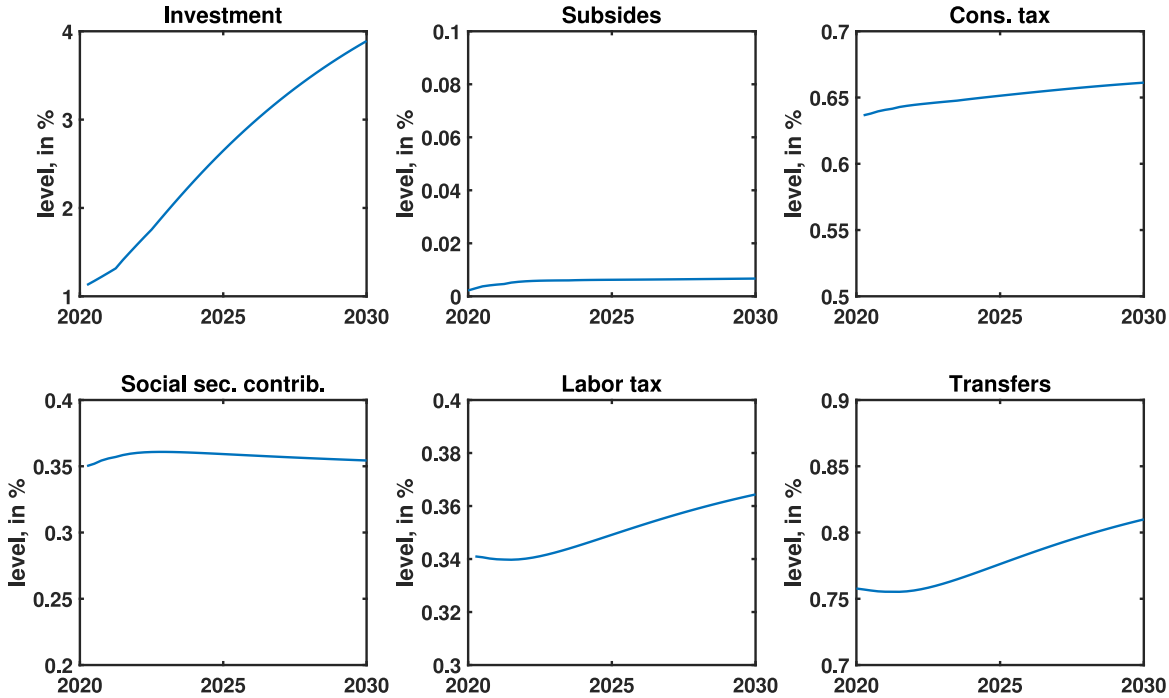

Fig. 6. Present value fiscal output multipliers across fiscal measures. ***Notes:*** Figure plots present value fiscal multipliers for output across fiscal instruments.

Table 2
Welfare assessment.

	Optimizer	Rule-of-thumb	Total
Baseline pandemic scenario	−1.12	−0.24	−0.72
...with fiscal intervention	−1.04	−0.16	−0.64

Notes: Welfare presented as life time consumption equivalents for different household types. Aggregate economy-wide welfare is calculated as $ce = (1-\mu)\cdot ce^o + \mu \cdot ce^r$.

(2008), among others, we see that the pandemic is a highly costly one-time event and that the huge fiscal stimulus package set up by the government noticeably alleviates these costs from an overall perspective.

6. Conclusion

The worldwide recession triggered by the COVID-19 pandemic required fiscal policy to step in to alleviate societal costs. The German government implemented various fiscal stabilization and stimulus measures. We assess this policy intervention by means of a New Keynesian dynamic stochastic general equilibrium model featuring a multi-sector production network, in which sectors are linked via intermediate-goods trade.

Our results indicate that the fiscal package indeed mitigates output and consumption losses notably in the short-run. The present value fiscal output multiplier of the package amounts to around 0.5 in the long-run, and the welfare costs of the pandemic can be reduced by around 11% on average, and even by 33% for liquidity-constrained households. The reduction of consumption taxation and direct transfers to households notably back consumption and stabilize output. Subsidies for firms are quite costly compared to their stabilizing effect, while public investment appears to be the most cost-effective instrument. However, the positive effects of the latter materialize in particular in the long-run.

Our paper sheds light on the question as to which fiscal policy measures help to mitigate the impact of the crisis and can provide guidance for possible further fiscal stimulus packages. Future research should address the long-term consequences of the pandemic and tools for fiscal policy to improve the long-run outlook as well as appropriate ways to ensure the sustainability of public finances.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.euroecorev.2023.104407.

References

Agnor, P.-R., Bratsiotis, G.J., Pfajfar, D., 2014. Credit frictions, collateral, and the cyclical behavior of the finance premium. Macroecon. Dyn. 18 (5), 985–997.
Arellano, C., Bai, Y., Mihalache, G., 2020. Deadly Debt Crises: COVID-19 in Emerging Markets. NBER Working Papers 27275.
Atalay, E., 2017. How important are sectoral shocks? Am. Econ. J.: Macroecon. 9 (4), 254–280.
Atkeson, A., 2020. What Will Be the Economic Impact of COVID-19 in the US? Rough Estimates of Disease Scenarios. NBER Working Papers 26867.
Attinasi, M.-G., Prammer, D., Stähler, N., Tasso, M., van Parys, S., 2019. Budget-neutral labor tax wedge reductions: A simulation-based analysis for the Euro area. Int. J. Central Bank. 15 (4), 1–54.
Auray, S., Eyquem, A., 2020. The macroeconomic effects of lockdown policies. J. Public Econ. 190, 104260.
Balleer, A., Zorn, P., Link, S., Menkhoff, M., 2020. Demand or Supply? Price Adjustment During the Covid-19 Pandemic. CEPR Discussion Papers 14907.
Baqaee, D.R., Farhi, E., 2022. Supply and demand in disaggregated keynesian economies with an application to the Covid-19 crisis. Amer. Econ. Rev. 112 (5), 1397–1436.
Bayer, C., Born, B., Luetticke, R., Müller, G., 2022. The coronavirus stimulus package: How large is the transfer multiplier? Econ. J. in press.
Benes, J., Kumhof, M., 2015. Risky bank lending and countercyclical capital buffers. J. Econom. Dynam. Control 58 (C), 58–80.
Berger, D., Herkenhoff, K., Huang, C., Mongey, S., 2022. Testing and reopening in an SEIR model. Rev. Econ. Dyn. 43, 1–21.
Bergholt, D., Sveen, T., 2014. Sectoral Interdependence and Business Cycle Synchronization in Small Open Economies. Working Paper 2014/04, Norges Bank.
Bom, P.R., Ligthart, J.E., 2014. What have we learned from three decades of research on the productivity of public capital? J. Econ. Surv. 28 (5), 889–916.
Born, B., Pfeifer, J., 2014. Risk matters: The real effects of volatility shocks: Comment. Amer. Econ. Rev. 104 (12), 4231–4239.
Boscá, J.E., Doménech, R., Ferri, J., García, J.R., Ulloa, C., 2021. The stabilizing effects of economic policies in Spain in times of COVID-19. Appl. Econ. Anal..
Bouakez, H., Rachedi, O., Santoro, E., 2018. Sectoral Heterogeneity, Production Networks, and the Effects of Government Spending. Cahiers de recherche 17–2018, Centre interuniversitaire de recherche en economie quantitative, CIREQ.
Bouakez, H., Rachedi, O., Santoro, E., 2021. The government spending multiplier in a multi-sector economy. Am. Econ. J.: Macroecon. forthcoming.
Boysen-Hogrefe, J., Fiedler, S., Groll, D., Jannsen, N., Kooths, S., Mösle, S., 2020. Deutsche Wirtschaft vor Mühsamer Erholung. Kieler Konjunkturberichte 68, Kiel Institute for the World Economy (IfW).
Brinca, P., Duarte, J.B., Faria-e Castro, M., 2021. Measuring sectoral supply and demand shocks during COVID-19. Eur. Econ. Rev. 139.
Brülhart, M., Trionfetti, F., 2001. Industrial specialisation and public procurement: Theory and empirical evidence. J. Econ. Integr. 16, 106–127.
Brülhart, M., Trionfetti, F., 2004. Public expenditure, international specialisation and agglomeration. Eur. Econ. Rev. 48 (4), 851–881.
Bundesbank, 2020a. Outlook for the German Economy for 2020 to 2022. Monthly Report June, Deutsche Bundesbank.
Bundesbank, 2020b. Public Finances. Monthly Report August, Deutsche Bundesbank.
Bundesbank, 2021. Public Finances. Monthly Report June, Deutsche Bundesbank.
Chari, V., Kirpalani, R., Phelan, C., 2020. The hammer and the scalpel: On the economics of indiscriminate versus targeted isolation policies during pandemics. Rev. Econ. Dyn. 42, 1–14.
Christiano, L.J., Eichenbaum, M., Evans, C.L., 2005. Nominal rigidities and the dynamic effects of a shock to monetary policy. J. Polit. Econ. 113 (1), 1–45.
Christl, M., Poli, S.D., Hufkens, T., Peichl, A., Ricci, M., 2022. The role of short-time work and discretionary policy measures in mitigating the effects of the covid-19 crisis in Germany. Int. Tax Public Finance.
Christopoulou, R., Vermeulen, P., 2012. Markups in the Euro area and the US over the period 1981–2004: a comparison of 50 sectors. Empir. Econ. 42 (1), 53–77.
Chudik, A., Mohaddes, K., Pesaran, M.H., Raissi, M., Rebucci, A., 2021. A counterfactual economic analysis of Covid-19 using a threshold augmented multi-country model. J. Int. Money Finance 119.
Coenen, G., Straub, R., Trabandt, M., 2013. Gauging the effects of fiscal stimulus packages in the euro area. J. Econom. Dynam. Control 37 (2), 367–386.
Cooley, T., Prescott, E., 1995. Economic growth and business cycles. In: Cooley, T. (Ed.), Frontiers in Business Cycle Research. Princeton University Press, Princeton University, NJ, pp. 1–38.
Darougheh, S., 2021. Dispersed consumption versus compressed output: Assessing the sectoral effects of a pandemic. J. Macroecon. 68, 103302.
Eichenbaum, M.S., Rebelo, S., Trabandt, M., 2021. The macroeconomics of epidemics. Rev. Financ. Stud. 34 (11), 5149–5187.
Elenev, V., Landvoigt, T., Van Nieuwerburgh, S., 2022. Can the covid bailouts save the economy? Econ. Policy 37 (110), 277–330.
Enders, A., Groll, D., Stähler, N., 2020. Parity funding of health care contributions in Germany: A DSGE perspective. Ger. Econ. Rev. 21 (2), 217–233.
Eurostat, 2008. NACE Rev. 2 – Statistical Classification of Economic Activities in the European Community. Methodologies and working papers.
Faria-e-Castro, M., 2021. Fiscal policy during a pandemic. J. Econom. Dynam. Control 125, 104088.
Forni, L., Monteforte, L., Sessa, L., 2009. The general equilibrium effects of fiscal policy: Estimates for the Euro area. J. Public Econ. 93 (3–4), 559–585.
Gadatsch, N., Hauzenberger, K., Stähler, N., 2016a. Fiscal policy during the crisis: A look on Germany and the Euro area with GEAR. Econ. Model. 52 (PB), 997–1016.
Gadatsch, N., Stähler, N., Weigert, B., 2016b. German labor market and fiscal reforms 1999–2008: Can they be blamed for intra-euro area imbalances? J. Macroecon. 50 (C), 307–324.
Gali, J., Gertler, M., Lopez-Salido, J.D., 2007a. Markups, gaps, and the welfare costs of business fluctuations. Rev. Econ. Stat. 89 (1), 44–59.
Gali, J., Lopez-Salido, J.D., Valles, J., 2007b. Understanding the effects of government spending on consumption. J. Eur. Econom. Assoc. 5 (1), 227–270.
GD, 2020. Erholung Verliert an Fahrt – Wirtschaft und Politik weiter im Zeichen der Pandemie. Gutachten vom 14, Gemeinschaftsdiagnose.
Gourinchas, P.O., Kalemli-Özcan, Ş., Penciakova, V., Sander, N., 2021. Fiscal Policy in the Age of COVID: 'Does it Get in all of the Cracks?'. NBER Working Papers 29293.
Greenwood, J., Hercowitz, Z., Huffman, G.W., 1988. Investment, capacity utilization, and the real business cycle. Amer. Econ. Rev. 78 (3), 402–417.

Guerrieri, V., Lorenzoni, G., Straub, L., Werning, I., 2022. Macroeconomic implications of COVID-19: Can negative supply shocks cause demand shortage? Amer. Econ. Rev. 112 (5), 1437–1474.
Hacıoğlu-Hoke, S., Känzig, D.R., Surico, P., 2021. The distributional impact of the pandemic. Eur. Econ. Rev. 134, 103680.
Hürtgen, P., 2020. Fiscal Sustainability During the COVID-19 Pandemic. Discussion Papers 35/2020, Deutsche Bundesbank.
Iacoviello, M., 2005. House prices, borrowing constraints, and monetary policy in the business cycle. Amer. Econ. Rev. 95 (3), 739–764.
Imrohoroglu, A., 2008. Welfare costs of business cycles. In: Macmillan, P. (Ed.), The New Palgrave Dictionary of Economics, Vol. 1, second ed. Palgrave Macmillan, London.
Ireland, P.N., 2003. Endogenous money or sticky prices? J. Monetary Econ. 50 (8), 1623–1648.
Jaimovich, N., Rebelo, S., 2009. Can news about the future drive the business cycle? Amer. Econ. Rev. 99 (4), 1097–1118.
Jordà, Ò., Singh, S.R., Taylor, A.M., 2022. Longer-run economic consequences of pandemics. Rev. Econ. Stat. 104 (1), 166–175.
Kaplan, G., Moll, B., Violante, G., 2020. The Great Lockdown and the Big Stimulus: Tracing the Pandemic Possibility Frontier for the U.S.. NBER Working Papers 27794.
Keen, B., Wang, Y., 2007. What is a realistic value for price adjustment costs in new keynesian models? Appl. Econ. Lett. 14 (11), 789–793.
King, R.G., Plosser, C.I., Rebelo, S.T., 1988. Production, growth and business cycles: I. The basic neoclassical model. J. Monetary Econ. 21 (2), 195–232.
Kirsanova, T., Wren-Lewis, S., 2012. Optimal fiscal feedback on debt in an economy with nominal rigidities. Econom. J. 122 (559), 238–264.
Klein, P., 2000. Using the generalized Schur form to solve a multivariate linear rational expectations model. J. Econom. Dynam. Control 24 (10), 1405–1423.
Leeper, E.M., Walker, T.B., Yang, S.-C.S., 2010. Government investment and fiscal stimulus. J. Monetary Econ. 57 (8), 1000–1012.
Mitchell, P.R., Sault, J.E., Wallis, K.F., 2000. Fiscal policy rules in macroeconomic models: principles and practice. Econ. Model. 17 (2), 171–193.
Pfeiffer, P., Roeger, W., in't Veld, J., 2020. The COVID19-Pandemic in the EU: Macroeconomic Transmission and Economic Policy Response. Discussion Paper 127, European Commission.
Priesmeier, C., Stähler, N., 2011. Long dark shadows or innovative spirits? The effects of (smoothing) business cycles on economic growth: A survey of the literature. J. Econ. Surv. 25 (5), 898–912.
Rotemberg, J.J., 1982. Sticky prices in the United States. J. Polit. Econ. 90 (6), 1187–1211.
Schmitt-Grohé, S., Uribe, M., 2007. Optimal simple and implementable monetary and fiscal rules. J. Monetary Econ. 54 (6), 1702–1725.
Smets, F., Wouters, R., 2003. An estimated dynamic stochastic general equilibrium model of the euro area. J. Eur. Econom. Assoc. 1 (5), 1123–1175.
Stähler, N., Thomas, C., 2012. FiMod - A DSGE model for fiscal policy simulations. Econ. Model. 29 (2), 239–261.
Timmer, M.P., Dietzenbacher, E., Los, B., Stehrer, R., De Vries, G.J., 2015. An illustrated user guide to the world input–output database: the case of global automotive production. Rev. Int. Econ. 23 (3), 575–605.
Trionfetti, F., 2000. Discriminatory public procurement and international trade. World Econ. 23 (1), 57–76.
Ziegelmeyer, M., Porpiglia, A., Teppa, F., Le Blanc, J., Zhu, J., 2015. Household Saving Behaviour and Credit Constraints in the Euro Area. Working Paper Series 1790, European Central Bank.